Einstein's solution: *There is no such thing as absolute space. There is no such thing as absolute time. Newton's foundation for all of physics was flawed And as for the aether: It does not exist.*

By rejecting absolute space, Einstein made absolutely meaningless the notion of "being at rest in absolute space." There is no way, he asserted, to ever measure the Earth's motion through absolute space, and that is why the Michelson–Morley experiment turned out the way it did. One can measure the Earth's velocity only *relative to other physical objects* such as the Sun or the Moon, just as one can measure a train's velocity only relative to physical objects such as the ground and the air. For neither Earth nor train nor anything else is there any standard of absolute motion; motion is purely "relative."

By rejecting absolute space, Einstein also rejected the notion that everyone, regardless of his or her motion, must agree on the length, height, and width of some table or train or any other object. On the contrary, Einstein insisted, *length, height, and width are "relative" concepts.* They depend on the relative motion of the object being measured and the person doing the measuring.

By rejecting absolute time, Einstein rejected the notion that everyone, regardless of his or her motion, must experience the flow of time in the same manner. *Time is relative,* Einstein asserted. Each person traveling in his or her own way must experience a different time flow than others, traveling differently.

It is hard not to feel queasy when presented with these assertions. If correct, not only do they cut the foundations out from under the entire edifice of Newtonian physical law, they also deprive us of our commonsense, everyday notions of space and time.

But Einstein was not just a destroyer. He was also a creator. He offered us a new foundation to replace the old, a foundation just as firm and, it has turned out, in far more perfect accord with the Universe.

Einstein's new foundation consisted of two new fundamental principles:

- *The principle of the absoluteness of the speed of light* Whatever might be their nature, space and time must be so constituted as to make the speed of light absolutely the same in all directions, and absolutely independent of the motion of the person who measures it.